PAPER #90 □ MUGE Compressed Gravity: Newtonian Base + 9 Corrections

Title: MUGE Compressed Gravity: A 10-Term Framework Correcting Newtonian Gravity at Galaxy-to-Cosmological Scales

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Framework: MUGE (Multi-Unit Gravity Expression), UQFF Star-Magic
Date: March 7, 2026
Source Data: validate_uqff_muge.py, source4.cpp (10 Compressed functions), compute_compressed_MUGE_SOURCE4
Index Slot: §1.12 UQFF Master Calculators, Paper #90

Abstract

The MUGE Compressed gravity framework extends Newtonian gravity with 9 physics-motivated corrections spanning expansion, magnetic suppression, envelope, Ug-sum, cosmological constant, quantum, fluid, and dark matter perturbation effects. `validate_uqff_muge.py` validates the complete 10-term sum for 5 astrophysical systems (Sgr A*, M87, Sun, NeutronStar, Magnetar), confirming no NaN/Inf across 8 distinct radial ranges and 5 mass scales.

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1. The 10-Term MUGE Compressed Framework

From `source4.cpp::compute_compressed_MUGE_SOURCE4`:

$$g_{\mathrm{MUGE}}^{\mathrm{Comp}}(r) = g_{\mathrm{Newton}} + \sum_{k=1}^{9} \delta_k(r)$$

Term Definitions

Term #	Symbol	Formula	Physics
0	g_Newton	GM/r^2	Base Newtonian gravity
1	d_Expansion	$H0^2 r/6$	Hubble flow correction
2	d_Super	$-B^2/(4p?\ r^2)$	Magnetic field suppression
3	d_Envelope	g_N × ?_env/?_crit	Gas envelope contribution
4	d_Ug_sum	$SUgk/(?\ r^3)$	UQFF Ug1+Ug2+Ug3+Ug4 sum
5	d_Cosm	$?c^2 r/3$	Cosmological constant (dark energy)
6	d_Quantum	$?^2/(M^2 r5)$	Quantum gravity correction
7	d_Fluid	$??^2 v/(?\ r)$	Navier-Stokes viscosity term
8	d_Perturbation	d_DM × g_N	Dark matter perturbation

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2. Term-by-Term Magnitudes at Sgr A* r_horizon = $1.27 \times 10^{1°}$ m

Term	Value at r_horizon	Relative to g_N
g_Newton	2.34×10^{2} m/s^{2}	1.000
d_Expansion	$7.8 \times 10?^{3}4$ m/s^{2}	$3.3 \times 10?^{3}6$
d_Super	$-1.2 \times 10?^{12}$ m/s^{2}	$-5.1 \times 10?^{1}5$
d_Envelope	$+8.5 \times 10?8$ m/s^{2}	$+3.6 \times 10?^{1°}$
d_Ug_sum	$+1.4 \times 10?6$ m/s^{2}	$+6.0 \times 10??$
d_Cosm	$-6.5 \times 10?^{2}6$ m/s^{2}	$-2.8 \times 10?^{2}8$
d_Quantum	$+3.2 \times 10?47$ m/s^{2}	$+1.4 \times 10?4?$
d_Fluid	$+7.6 \times 10?^{1}?$ m/s^{2}	$+3.2 \times 10?^{21}$
d_Perturbation	$+4.7 \times 10?4$ m/s^{2}	$+2.0 \times 10?6$
g_total	**2.340×10^{2} m/s^{2}**	**1.000002**

No NaN/Inf □ PASS. Total correction relative to Newton: < 5 ppm at r_horizon.

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3. Dominant Corrections by Scale

Galactic Scale (r ~ kpc = $3 \times 10^{1}?$ m)

At galactic radius r = 1 kpc from Sgr A*:

Dominant corrections	Relative magnitude
d_DM Perturbation	~$10?^{1}$ (DM halo)
d_Expansion	~10?5 (sub-dominant at kpc)
d_Ug_sum	~10?7

? Dark matter perturbation dominates at galaxy scales. MUGE compressed reduces to DM+Newton.

Cosmological Scale (r ~ Gpc)

Dominant corrections	Relative magnitude
d_Expansion	~$10?^{1}$ (Hubble flow)
d_Cosm	~$10?^{1}$ (dark energy)

? Expansion and ? dominate at Gpc. MUGE compressed ? ?CDM-concordant.

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4. Cross-System Validation (validate_uqff_muge.py)

All 5 systems from validator, all 10 MUGE terms verified finite:

System	M (kg)	r_test (m)	g_MUGE (m/s²)	NaN/Inf
Sgr A*	8.0×10^{36}	1.27×10^{10}	234.3	None
M87*	1.26×10^{40}	1.95×10^{13}	2.21×10^{3}	None
Sun	1.99×10^{30}	6.96×10^{8}	274.3	None
NeutronStar	2.8×10^{30}	1.2×10^{4}	1.62×10^{12}	None
Magnetar	3.0×10^{30}	1.2×10^{4}	1.74×10^{12}	None

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5. Relationship to UQFF_CompressedCalculator

The `UQFF_CompressedCalculator` (Paper #89) wraps the MUGE Compressed result and adds the d_Ug factor:

$$F_{\text{Comp}}^{\text{UQFF}}(r,t) = g_{\text{MUGE}}^{\text{Comp}}(r) + \delta_{\text{Ug}}(r,t)$$

Where d_Ug includes all 4 Ugk terms evaluated in the UQFF framework (not just their sum).

Summary

Scale	Dominant correction(s)	MUGE expansion
Near-horizon	d_Ug_sum + d_Perturbation	UQFF » GR
Galactic (kpc)	d_DM (×0.1 g_N)	Dark matter-driven
Cosmological (Gpc)	d_Expansion + d_Cosm	?CDM-concordant
All scales	No NaN/Inf (5 systems)	Numerically stable

Source: validate_uqff_muge.py | source4.cpp compute_compressed_MUGE_SOURCE4 | 5 systems × 10 terms all finite

See also: PAPER_089 | Part of the Star-Magic UQFF Whitepaper Series.

UQFF computed: MUGE buoyancy ratio U_bi/F_U = [SSq]×?×r²/GM = 5.7e-1×5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s² at r_ISCO.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
Îº	5.0 Ã 10â»â´ dayâ»Â¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
Î²_i	0.60â0.61	Buoyancy coupling coefficient
kâ	1.5	Ug1 DPM-dipole coupling
kâ	1.2	Ug2 outer-bubble charge coupling
kâ	1.8	Ug3 string-rotation coupling
kâ	2.0	Ug4 vacuum-concentration coupling
Î·	10â»Â²Â²	Inertia tensor scale
E_react(0)	10â´â¶ J	Reference reactive energy

A.2 F_U Master Equation (Complete â 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
âÎ£Î»áµ¢Â·Uáµ¢Â·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
Î»â=10â»Â¹â°, Î»â=10â»Â¹Â², Î»â=10â»Â¹Â¹, Î»â=10â»Â¹Â³ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q \cos(\Delta\omega\, t)igr)$$

Symbol	Value	Description
Ï□_c	10Â¹â□µ kg/mÂ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Î□Ï□	2Ï□/(434Â·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, â□¦)	Multi-scale field interactions
Buoyant	Î²_i Ã□ Ubi	Expanding nebulae, stellar winds
Superconductive	Um Ã□ (1+10Â¹Â³Â·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.